

Informed Options Trading Prior to Takeover Announcements: Insider Trading?*

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Abstract

We quantify the pervasiveness of informed trading activity in target companies' equity options before the announcements of 1,859 U.S. takeovers between 1996 and 2012. About 25% of all takeovers have positive abnormal volumes, which are greater for short-dated out-of-the-money calls, consistent with bullish directional trading before the announcement. Over half of this abnormal activity is unlikely due to speculation, news and rumors, trading by corporate insiders, leakage in the stock market, deal predictability, or beneficial ownership filings by activist investors. We also examine the characteristics of option trades litigated by the SEC for alleged illegal insider trading. While the characteristics of such trades closely resemble the patterns of abnormal option volume in the U.S. takeover sample, we find that the SEC litigates only about 8% of all deals in it.

Keywords: Civil Litigations, Derivatives, Insider Trading, Mergers and Acquisitions, SEC

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1 Introduction

How pervasive is informed options trading around takeover announcements? Cao et al. (2005), for example, suggest that abnormal trading in equity options prior to takeover announcements is informed, as directional trading activity positively foreshadows future price movements.¹ This finding raises questions about the nature of the run-up in option volumes before takeover announcements and the distribution of informed trading across deals. Is the run-up driven by a few deals with significant volumes, or do all deals contribute to the options activity that is abnormal on average? What information drives the individual run-ups in option volumes, and can they be explained by publicly available sources of information besides illegal insider trading? In this paper, we characterize the pervasiveness of informed options trading around takeover announcements and study the sources of the pre-announcement run-up in option volumes.

We first quantify the pervasiveness of informed trading in the equity options of target companies ahead of 1,859 U.S. takeover transactions between January 1996 and December 2012. To better understand the sources of informed trading, in the second step, we examine a large number of channels that could plausibly explain the abnormal trading volumes in options based on correct anticipation of upcoming takeover activity. It is the deal-by-deal examination of the pre-announcement activity that allows us to examine the overlaps in explanations across deals, and to identify the deals with abnormal options activity that are unlikely to be anticipated. Third, we also verify whether the run-up may be the result of illegal insider trades reported by the Securities and Exchange Commission (SEC), and compare the characteristics of all option trades litigated for alleged illegal insider trading to those of the pre-announcement options activity. This comparison strengthens our assessment of the nature and sources of the pre-announcement run-up in option volumes.

We document that about 25% (467) of all deals in our sample have abnormal volumes of equity options, over the 30-day period preceding the takeovers, which are statistically significant at the 5% level. The proportion of cases with abnormal volumes is relatively higher for call options than for put options. Stratifying the results by “moneyness,” we find that there is significantly higher abnormal trading volume in out-of-the-money (OTM) call options compared to at-the-money (ATM) and in-the-money (ITM) calls.²

¹We discuss other references on informed options trading prior to takeover announcements in Section 2 and summarize them in Table 1.

²We also find that ITM puts trade in abnormally larger volumes than ATM puts. This indicates that informed

An examination of the characteristics of cumulative abnormal volume shows that informed trading is more pervasive for larger deals, those for which informed investors may potentially have less uncertainty about the final takeover price, and in cases where target firms receive cash offers. We study the trading volume, implied volatility, and bid-ask spreads of equity options, and consider a number of robustness tests, which support the evidence of informed investors trading directionally in anticipation of a price jump in the target company's stock. It is the deal-by-deal examination of the pre-announcement activity that helps us appreciate the pervasiveness of informed trading in takeover transactions.

We next explore whether the takeover announcements could have been anticipated based on public sources of information. We first show that the run-up in options volume is unlikely due to speculative trading activity in response to observable stock trading activity or industry and firm characteristics. We compare the options activity in the takeover sample to several control samples that are matched either on the stock market activity, or on industry and firm characteristics. Informed trading activity in options ahead of takeover announcements is absent from these control groups.

In addition, we examine how much of the informed options activity can be explained by news and rumors. To identify rumors and news about upcoming takeovers, we use RavenPack News Analytics, a database that is constructed from textual information in major newspaper outlets, public relations feeds, and over 19,000 other traditional and social media web sites. We associate news with 170 takeover transactions, and with only 9% (40/467) of the deals that exhibit informed options activity. We find no statistically significant difference in the average cumulative abnormal options trading volumes between the sample of 170 takeovers with news and the sample of 1,689 without news. We further check whether the option trades originate from corporate insider accounts. Corporate officers, directors, or large block-holders are legally required to disclose security transactions in their company's options. An analysis of the derivative transactions and holdings information in the Thomson Reuters insider filings reveals that not a single options transaction was executed by registered insiders in the 30-day period before the announcement.

We also consider the possibility that astute option traders trade on information leakage in the stock market. However, our analysis suggests that option volume leads stock volume, and that past traders may possibly also engage in ITM put transactions, or that the call trading generates arbitrage-based put trading activity. We explicitly consider synthetic options strategies and show in an Online Appendix (Section A-I) that a wide variety of strategies for exploiting private information about an acquisition result in the trading of OTM calls or ITM puts.

stock volume and return performance is not significantly related to future abnormal options activity. In addition, we find that only 7% of the deals in our sample exhibit abnormal stock returns in the pre-announcement period, while about 44% of all deals exhibit excess implied volatility. Although 19% of the takeover transactions exhibit statistically significant abnormal stock volume, a frequency somewhat lower than in the options market, the economic magnitude is substantially smaller. Thus, quantifying how many deals are subject to informed trading may also be informative about whether informed trading is more prevalent in the options or in the stock market.

Next, we show that it is difficult to predict takeover announcements based on publicly available information. Thus, the documented abnormal option trading volume is unlikely due to traders' ability to time the market. However, we observe that most of the informed activity arises in the five to ten days before the information gets publicly released. Finally, we screen the 13D beneficial ownership reports, which need to be filed by registered active investment advisors no later than 10 days following the acquisition of beneficial ownership of more than 5% of any class of publicly traded securities. The trading we identify by supposedly informed activist investors is unlikely to fully explain the abnormal options activity, as only 17 of these deals have a filing in the 30-day period prior to the takeover announcement date.

Out of the 467 deals in our sample with significant cumulative abnormal options volume, 13% (236) are unlikely to be associated with publicly available sources of information. In the sub-sample of 236 deals, we exclude those deals that exhibit statistically significant stock volume or returns, which appear to lag the option market in the takeover pre-announcement period. An additional channel that may drive the run-up in options volume could be illegal insider trading. Therefore, we filter through more than 8,000 public SEC litigation records dated between 1990 and 2013 to identify whether the takeover transactions in our sample were subject to a litigation for alleged insider trading. We find that the SEC litigated about 8% of the takeovers in our sample for insider trading in options or stocks, and only 9% (43/467) of the transactions that we associate with informed trading. Moreover, only 10% (24/236) of the deals that we fail to associate with public sources of information are involved in a litigation. Thus, the number of civil lawsuits for insider trading appears modest in comparison to the pervasive informed trading activity reflected in 25% of the takeover transactions.

Using the SEC's litigation records, we hand-collect information on the size, timing, and type of

illegal trades; we supplement with information in the criminal records from the U.S. Department of Justice (DoJ). The characteristics of the illegal option trades, i.e., short-dated OTM call options on target companies that are initiated, on average, 21 days before the announcement, closely resemble the characteristics and timing of the abnormal options activity in a representative sample of takeover transactions. This resemblance, coupled with the absence of public information sources that could have led to the anticipation of the takeover transaction, calls for further regulatory examination of the informed options activity.

DeMarzo et al. (1998) suggest that it may be optimal to prosecute insiders only after large price moves or after large volume transactions. We find that the SEC is likely to examine large target firms that experience substantial abnormal returns after the takeover announcement, and in which the acquirers are headquartered outside the U.S. We find, however, no evidence that the probability of litigation is positively related to the pre-announcement abnormal options volume.

We extend the literature along three key dimensions. In this paper, we quantify the prevalence of informed trading and document that the informed options activity is driven by 25% of all deals, in contrast to prior research. Second, we diverge from previous work and examine the sources of informed trading in the options market and show that for at least 13% of all deals, it is difficult to associate the abnormal options activity with public sources of information. Third, a unique feature of our research is that we study the characteristics of SEC-litigated cases related to insider trading in options prior to takeover announcements. This allows us to compare the nature of abnormal options activity to illegal option trades and examine how the SEC's litigation record relates to abnormal options trading around takeover announcements.

2 Literature Review

The run-up in the stock prices of target companies before M&A announcements is well-documented (Mandelker; 1974; Dodd; 1980; Jensen and Ruback; 1983; Asquith; 1983; Dennis and McConnell; 1986; Schwert; 1996). A long-standing debate relates to whether this run-up is due to public information such as, for example, media speculation (Jarrell and Poulsen; 1989), or whether it is the result of private information leakage and illegal insider trading (Keown and Pinkerton; 1981; Meulbroek; 1992; Sanders and Zdanowicz; 1992).

Abnormal options volume and price activity ahead of M&A announcements have been the subject of many papers, particularly in recent years. Jayaraman et al. (1991) are the first to document a pre-announcement increase in the option-implied volatilities of target companies, that precede an increase in stock returns, as confirmed by Levy and Yoder (1993). Jayaraman et al. (2001) document that the abnormal options volume is accompanied by abnormal open interest that is concentrated in short-term OTM options, that it leads abnormal stock volume, and that the increase in abnormal options volume is greater for call than for put options. Using signed volume, Cao et al. (2005) support the presence of informed activity, since prior to takeover announcements, the option volume order imbalance contains information regarding subsequent stock price movements, greater call volume balances are associated with greater announcement returns, and the options market displaces the stock market for information-based trading (see also Arnold et al. (2006)). Acharya and Johnson (2010) show that a large number of equity participants in leveraged buyout syndicates is associated with greater levels of suspicious stock and options activity. Clements and Singh (2011) argue that pre-announcement options volume reflects both informed and “contraire” trading, while Shafer (2012) documents a positive correlation between the pre-announcement option-to-stock volume ratio and the probability of being a takeover target that becomes weaker after Regulation Fair Disclosure in 2000.

Many studies have corroborated the evidence of abnormal pre-announcement option activity in target companies in the U.S. and U.K. (Spyrou et al.; 2011). The literature emphasizes different aspects of the pre-announcement activity, including abnormal changes in the IV skew, IV spread, and the option-to-stock volume ratio (Klapper; 2013), a positive correlation between pre-announcement run-up and abnormal options volume (Wang; 2013), an increasing importance of options’ leading role for price discovery (Liu et al.; 2015), or a greater propensity of informed trading to occur in more liquid and higher leverage options (Podolski et al.; 2013). Chan et al. (2015) show that the one-day pre-event implied volatility spread (the implied volatility skew), a proxy for informed option trading, is positively (negatively) associated with acquirer cumulative abnormal returns. Ordu and Schweizer (2015) associate greater abnormal volumes with greater CEO wealth-to-performance sensitivity for stock-financed takeovers, suggesting that informed managers hedge anticipated negative acquirer announcement returns. Focusing also on acquirers, Huang and Tung (2016) find a positive relation between announcement returns and pre-announcement option-to-stock volume ratios, which

are positively related to idiosyncratic stock volatility. Chesney et al. (2015) propose a method for detecting abnormal options activity and relate six unusual transactions to M&A announcements. Kedia and Zhou (2014) conclude in favor of pre-announcement informed trading in target bonds. Poteshman (2006) concludes that informed investors traded put options ahead of the 9/11 terrorist attacks.

In Table 1, we show the differences between this study and prior work. Apart from Spyrou et al. (2011), who examine U.K. data, most studies are not informative about the distribution of informed options trading. For a small sample that is not representative of option trading over the last two decades, Jayaraman et al. (1991) and Jayaraman et al. (2001) identify positive changes in implied volatilities, without mentioning statistical significance. In contrast, we examine deals, case by case, and emphasize how and where insiders trade in the options market: they engage in directional strategies for targets, which are reflected in more pronounced abnormal activity in OTM calls and cash-financed takeovers. In addition, we consider synthetic option strategies that lead to long bullish or short bearish exposures for targets, and review earlier evidence as the findings appear to be inconsistent across studies.³

Importantly, there is only scarce information on the sources of informed options trading for target firms. Klapper (2013) and Wang (2013) examine media speculation and rumors, but their results are inconsistent with each other.⁴ Liu et al. (2015) provide some evidence against the leakage hypothesis, but do not explain whether abnormal options activity for a given deal might be the result of abnormal stock activity. Lowry et al. (2016) indicate that trading desks connected to M&A advisory desks take abnormal call option positions in the target companies from seven quarters to one quarter before the announcement. Ordu and Schweizer (2015) examine the sources of informed trading (managers), but focus on the options of the acquirers. Our focus here is on documenting the abnormal options activity for the target firms and identifying the specific types of options traded before the announcement.

Some authors examine the illegal insider trading investigated by the SEC (Jayaraman et al.; 2001; Podolski et al.; 2013) or study the predictability of SEC litigations (Wang; 2013). Using a

³For instance, Cao et al. (2005) and Wang (2013) document greater abnormal options activity in short-term OTM and ATM options, respectively, while Chesney et al. (2015) find that abnormal volumes are greater for put options.

⁴Wang (2013) finds no significant difference in abnormal options volume for samples with and without media coverage. Klapper (2013) finds a greater stock price run-up for rumored deals. Again, in contrast we use a novel high frequency news data base to revisit the evidence, and quantify the fraction of rumored deals.

proprietary sample of illegal trades litigated by the SEC, Meulbroek (1992) and Meulbroek and Hart (1997) document that insider trades have an immediate effect on stock prices, that half of the pre-announcement run-up occurs on insider trading days, and that the announcement returns are a third larger when insider trading is detected. Frino et al. (2013) study hand-collected SEC litigations and find that illegal stock trades are positively associated with subsequent price changes, but negatively associated with the size of the penalties and the stock's liquidity. However, we find no information in the literature on the characteristics and patterns of option trades litigated for illegal insider trading, nor on how prosecution relates to the degree of abnormal options activity. In this paper, we relate our results to the literature associated with illegal insider trading in stocks.

Cornell and Sirri (1992) and Chakravarty and McConnell (1999, 1997) conduct clinical studies of illegal stock trades ahead of the 1982 takeover of Campbell Taggart by Anheuser-Busch and the 1984 takeover of Carnation by Nestlé. Both studies find positive price impacts, and either a positive or no effect on bid-ask spreads or depth. Fishe and Robe (2004) find that trading by brokers, who illegally had advance access to news information on 116 stocks, negatively impacted market depth. Guercio et al. (2017) argue that illegal insider trading has decreased in response to more aggressive regulatory enforcement. Ahern (2017) describes insider trading networks from civil and criminal prosecutions initiated by the SEC and the DoJ, while Kacperczyk and Pagnotta (2016) show that market price and liquidity signals are impacted by illegal trading using SEC litigation information. Heitzman and Klasa (2016) document that investors trade quickly in stocks upon new information from non-public merger negotiations. Bhattacharya (2014) provides a comprehensive literature review of insider trading. For a discussion on the legal aspects of insider trading, see, for example, Arshadi (1998) and Crimmins (2013).

Our focus differs from the literature examining informed stock trading by corporate insiders. For example, Cohen et al. (2012) show that only opportunistic transactions, as opposed to routine transactions, have predictive content for stock prices. Agrawal and Nasser (2012) discuss widespread “passive” insider trading on targets, whereby registered insiders increase their net exposure through reduced stock selling. Our focus is on option trading, not stock trading.

Our work broadly relates to the literature on when and how informed agents choose to trade in the options market in the presence of asymmetric information (Easley et al.; 1998), differences in opinion (Cao and Ou-Yang; 2009), short-sale constraints (Johnson and So; 2012), or margin requirements

and wealth constraints (John et al.; 2003), and on the predictability of option-implied measures for stock returns (Easley et al.; 1998; Pan and Poteshman; 2006; Cremers and Weinbaum; 2010; Xing et al.; 2010; Johnson and So; 2012; Jin et al.; 2012; Hu; 2014; Tse-Chun and Xiaolong; 2015). Several other papers are peripherally related to this paper. Roll et al. (2010) study the relation between the option-to-stock trading volume and post-earnings announcement returns. Subramanian (2004) and Bester et al. (2011) develop theoretical option pricing models for the target in the case of stock-for-stock and cash mergers respectively.

3 Data Selection and Takeover Deal Characteristics

The data for our study come from three primary sources: the Thomson Reuters Securities Data Company Platinum (SDC) database, the Center for Research in Securities Prices (CRSP) database and the OptionMetrics database. We start our sample selection with the full domestic M&A dataset for U.S. target firms from SDC Platinum for the time period from January 1996, the starting date for available option information in OptionMetrics, through December 2012. Our final sample consists of 1,859 takeover transactions for which we were able to identify matching stock and option information for the target. These deals were undertaken by 1,279 unique acquirers on 1,669 unique targets.⁵

Starting from an initial sample of 185,419 transactions, we restrict the study to deals aimed at affecting a change of control, where the acquirer owned less than 50% of the target's stock before the transaction, and was seeking to own more than 50% after it. Hence, our sample includes only M&As of majority interest, excluding all deals that were acquisitions of remaining or partial interest (minority stake purchases), acquisitions of assets, recapitalizations, or buybacks/repurchases/self-tender and exchange offers. In addition, we exclude deals with pending or unknown status (i.e., we only include completed, tentative or withdrawn deals). These restrictions reduce the sample size to 34,350 deals. Next, we require information on the deal value, and eliminate deals with a transaction value below 1 million USD, which reduces the sample further to 19,064 transactions. Finally, we match the information from SDC with price and volume information in both the CRSP and OptionMetrics databases. We require a minimum of 90 days of valid stock and option price and volume information on the target prior to, and including, the announcement date, which results in the final full sample of

⁵Thus, 190 of the targets were involved in an unsuccessful merger or acquisition that was ultimately withdrawn. However, we include these cases in our sample, since the withdrawal occurred after the takeover announcement.

1,859 takeover announcements. All matches between SDC and CRSP-OptionMetrics are manually checked for consistency based on the company name.

Panel A in Table 2 reports the basic deal characteristics for the full sample. Pure cash offers make up the largest fraction of the sample (48.6%), followed by hybrid financing offers (22.3%) and share offers (21.7%). Only 82.9% of all transactions are completed, and mergers tend to be mostly within the same industry, with 53.4% of all deals being undertaken with a company in the same industry based on the two-digit SIC code. We find that 90.2% of all deals are considered to be friendly and only 3.4% are hostile, while 11.6% of all transactions are challenged. For only 6.5% of the sample do the contracts contain a collar structure, 76.5% of all deals involve a termination fee, and in only 3.5% of the transactions does the bidder already have a toehold in the target company. Panel B shows that the average deal size is 3.8 billion USD, with cash-only deals being, on average, smaller (2.2 billion USD) than stock-only transactions (5.4 billion USD). The average one-day offer premium, defined as the excess of the offer price relative to the target's closing stock price, one day before the announcement date, is 31%.

4 Informed Options Activity Prior to Takeovers

The first objective of our empirical analysis is to quantify the prevalence of informed trading using options volume. Investors with private information about the anticipated announcement return trade off the options' market leverage (Black; 1975) against the greater stock market liquidity and perhaps a lower probability of detection. In the presence of asymmetric information (Easley et al.; 1998), wealth constraints (John et al.; 2003), short-sale constraints (Johnson and So; 2012), or disagreement (Cao and Ou-Yang; 2009), some informed investors will migrate towards the option market. A necessary condition for informed pre-announcement activity is, therefore, the detection of abnormal options volume, as stated in hypothesis H1.

- H1: *There is evidence of positive abnormal trading volume in the equity options of target firms prior to takeover announcements.*

Investors with inside information would pursue directional trading strategies on the target firm's stock, as it almost always goes up after a takeover announcement (Andrade et al.; 2001). Thus, in the presence of superior information, a trading strategy involving the purchase of OTM call options

should generate a significantly higher abnormal return, as a consequence of the higher leverage (“more bang for the buck”). Hence, we expect a larger increase in abnormal trading volume for OTM calls relative to ATM and ITM calls. Moreover, an informed investor, taking advantage of his privileged knowledge of the future direction of the target’s stock price, may also increase the trading volume through the sale of ITM puts, as these will become less valuable with the increase in the target’s stock price upon announcement. An alternative, and more cash-intensive, strategy would be to mimic the strategy of buying OTM calls, by buying ITM puts coupled with the underlying stock, financed by borrowing. Thus, an abnormally high volume in ITM puts may result from either mimicking the purchase of OTM calls, or taking a synthetic long position in the stock (buying a call and selling a put with the same strike price). An informed trader may possibly engage in more complicated trading strategies to hide his intentions. Irrespective of which strategy is employed, we should observe abnormal trading volume in OTM call and/or ITM put options, if investors with precise information exploit option leverage. This leads to a sharper prediction in hypothesis H2.

- H2: *The ratios of the abnormal trading volumes in (a) OTM call options to ATM and ITM call options, and (b) ITM put options to ATM and OTM put options, written on the target firms, are higher prior to takeover announcements.*

To test hypotheses H1 and H2, we examine the deal-by-deal trading volume in equity options written on target firms during the 30 days preceding takeover announcements. In a nutshell, we find that approximately 25% of all deals in our sample exhibit statistically significant abnormal options activity (at the 5% level) in the pre-announcement period. In the U.K., Spyrou et al. (2011) also document abnormal pre-announcement options volume for 25% of the deals. The magnitude of abnormal volume is greater for OTM call options than for ATM and ITM calls in our sample, confirming the results of Cao et al. (2005). Our results suggest that the odds of abnormal volumes being greater in a sample with randomized announcement dates are at most one in a million.

4.1 Identifying Abnormal Trading Volumes

We test hypothesis H1 by applying event study methodology to trading volumes. To compute the abnormal trading volume, we use, as a conservative benchmark, a market model for volume (*MMV* model), which accounts for the market volume in options (median trading volume across all options

in the OptionMetrics database), the Chicago Board of Options Exchange (CBOE) Volatility Index (VIX), as well as the contemporaneous return of the underlying stock and the market, proxied by the return on the S&P 500 Index. In addition, we control for lagged values of the dependent and all independent variables. The estimation window starts 90 days before the announcement date and finishes 30 days before the announcement date. As we are interested in the abnormal trading volume in anticipation of the event, we use a 30-day event window before the announcement date. To account for the possibility of clustered event dates, we correct standard errors in aggregate tests for cross-sectional dependence.

In Panel A of Table 3, we show that the average cumulative abnormal trading volume for target firms is positive and statistically significant. The magnitude of the average cumulative abnormal volume over the 30 pre-event days is estimated to be 8,946 contracts for call options. For put options on the target, the average cumulative abnormal volume is also positive, but, over the 30 pre-event days, it is much smaller, at 1,559 contracts, and not statistically significant. The evolution of the average abnormal and cumulative abnormal trading volume for the targets is illustrated in Figure 1. It is apparent that the average cumulative abnormal trading volume in put options is quantitatively less important than that in call options, which primarily drives the results for the overall sample. The daily average abnormal volume for call options is positive and increases to approximately 1,500 contracts the day before the announcement. Individually, the number of deals with positive abnormal trading volumes, at the 5% significance level, ranges from 467 for calls, to 304 for puts, corresponding to approximately 25% and 16% of the entire sample, respectively.⁶ Thus, approximately 25% of the deals exhibit statistically significant cumulative abnormal options trading volume.⁷

We further stratify our sample by moneyness, and conduct an event study for each category, using only options expiring after the announcement date. We find that there is significantly higher abnormal trading volume for the targets in OTM call options, compared to ATM and ITM calls,

⁶In untabulated results, we find that, at the 1% significance level, the number of deals with positive abnormal trading volumes in the entire sample ranges from 275 for calls to 179 for puts, corresponding to frequencies of 15% and 10%, respectively.

⁷For statistical inference, we follow Campbell et al. (1996) and Kothari and Warner (2007). We note that the predictive volume models account for lagged values of both dependent and independent variables in order to purge serial correlation in the residuals at the firm level. Moreover, we show in the Online Appendix that the effects for option volumes do not arise in samples matched on randomized announcement dates, on industry and firm characteristics, as well as on takeover propensity scores. We also account for false positives in multiple hypothesis testing, following the methodology in Barras et al. (2010). The frequency of mergers with statistically significant abnormal call option volume is still 24%, even though we fail to adjust for false negatives, i.e., the fact that we fail to reject the null hypothesis simply by chance, which leads to a downward bias in the proportion of anomalous trading we observe.

both in terms of volume levels and frequencies. Table 3 shows that the average cumulative abnormal volume is 3,380 (1,417) contracts for OTM calls (puts) and 1,540 (984) contracts for the ITM calls (puts), while it is 1,156 (457) for ATM calls (puts). These values correspond to 408 (343, 482) deals, or 22% (18%, 26%) of the sample for OTM (ATM, ITM) calls, and 451 (362, 396) deals or 24% (19%, 21%), for OTM (ATM, ITM) puts, respectively.

In Panel B of Table 3, we report results from paired *t*-tests for the differences in the means of the cumulative average abnormal volumes across different categories. Consistent with hypothesis H2, these results emphasize that there is higher abnormal trading volume for OTM call options than for ATM or ITM calls. The differences in the means for OTM calls relative to ATM and ITM calls, are 2,224 and 1,840, respectively, which are positive and statistically different from zero. On the other hand, the difference in the means between ATM and ITM calls is slightly negative (-384), but not statistically different from zero. The average cumulative abnormal volume for ITM put options is higher than for ATM put options, which provides some evidence that informed traders may not only engage in OTM call transactions but may also sell ITM puts.⁸ The weaker evidence that informed traders may also engage in writing ITM put options may be due to the riskiness of writing naked (especially ITM) puts, as the failure of deal negotiations could lead to a sharp stock price drop. Selling naked puts also requires large margins, which may impose binding capital constraints on traders.

We verify our results using alternative tests and robustness checks. The results agree with the previous findings, yielding either similar or stronger results, both qualitatively and quantitatively. We discuss the details of these additional tests in the Online Appendix Section A-II.

4.2 Characteristics of Abnormal Volume

We next examine whether certain target companies are more likely than others to exhibit unusual trading volumes. We investigate several takeover deal characteristics that may imply a higher likelihood of informed trading, as they are associated with greater announcement returns. Hence, we regress the cumulative abnormal log trading volume in call and put options over the 30 pre-announcement days on a set of categorical variables reflecting deal characteristics and market activity

⁸The expected cumulative abnormal volume for OTM put options is slightly higher than that for ITM put options. The difference of 433 contracts is, nevertheless, small in unit terms, given that it is a cumulative measure taken over 30 days.

variables. We first test the following model:

$$\begin{aligned} CABVOL = & \beta_0 + \beta_1 SIZE + \beta_2 CASH + \beta_3 TOE + \beta_4 PRIVATE + \beta_5 COLLAR \\ & + \beta_6 TERM + \beta_7 FRIENDLY + \beta_8 US + \gamma_t + \varepsilon, \end{aligned} \quad (1)$$

where *CABVOL* denotes the cumulative abnormal trading volume in call or put options respectively, which we scale for each target by the average predicted volume in the event window.⁹ All specifications contain year fixed effects γ_t , and standard errors are clustered by announcement day.

Our strongest prior is that cumulative abnormal volume should be higher for cash-financed deals, given that they are known to have higher abnormal announcement returns (Andrade et al.; 2001) and are more likely to be completed (Fishman; 1989). Thus, we expect that an informed trader will benefit more from trading if he anticipates a higher abnormal return and is more certain that he will earn it. We test for this by including a dummy variable *CASH* that takes the value one for cash-financed deals. In addition, traders with private information may prefer opening positions for larger companies, whose stocks (and, therefore, their options) tend to be more liquid, and hence less likely to reveal unusual, informed trading. Thus, we expect cumulative abnormal volume to be higher for larger deals, measured by *SIZE*, a dummy variable that takes the value one, if the deal is above the median transaction value, and zero otherwise. According to Acharya and Johnson (2010), firm size may also proxy for the number of insiders to the deal and, thus, the probability of information leakage. A bidder who has a toehold in the company (*TOE*) may also be likely to gather and trade on private information about a future takeover. Alternatively, a toehold investor with privileged information may refrain from trading as he would be among the first suspects in any investigation. A foothold may also be interpreted as publicly observable information (Jarrell and Poulsen; 1989). We also control for other deal characteristics, such as whether the target is taken private post-takeover (*PRIVATE*), whether the deal has a collar structure (*COLLAR*), whether it involves a termination fee upon the failure of deal negotiations (*TERM*), whether the deal attitude is considered to be friendly (*FRIENDLY*), and whether the bidder is a U.S.-headquartered company (*US*).

The results for the benchmark regressions of cumulative abnormal volume in the target call options are reported in column (1) of Table 4. We find that firm size is a significant positive

⁹This analysis is based on a natural log transformation of volume. Hence, scaled cumulative abnormal log volume is comparable across companies, and interpretable as a percentage relative to predicted volume.

predictor of abnormal options volume. This evidence is consistent with the view that informed trading in target call options is greater for larger, more liquid companies, for which it is easier to hide informed trading, and when there is a greater probability of information leakage (Acharya and Johnson; 2010). Our results are similar if we proxy firm size using company sales. We also examined the number of target and acquirer advisors as a proxy for information leakage, but the results are not statistically significant, similar to the findings of Heitzman and Klasa (2016). This may be due to insufficient variation in the number of advisors for publicly traded firms (91.2% of the sample has less than 5 advisors, and the median is 2), whereas Acharya and Johnson (2010) examine a sample of private equity buyouts, which typically feature a higher number of equity participants. Quantitatively, we find that a target deal above the median transaction value has, on average, 3.32% greater cumulative abnormal call trading volume relative to its normal volume than a target below the median deal size.

The results in column (1) of Table 4 suggest that cash-financed deals have, on average, 6.37% greater cumulative abnormal volume than non-cash-financed deals. Given that the average cumulative abnormal volume is approximately 10,000 contracts, the typical cash-financed deal has about 637 more contracts traded during the 30-day pre-announcement period. The cash indicator is consistently robust across all specifications, with similar economic magnitudes.

If the bidder already has a toehold in the company, cumulative abnormal volume is about 5.6% smaller. The negative coefficient supports the interpretation that investors with a toehold may make more of an attempt to keep their intentions secret, as they would be natural suspects in the case of insider trading. The coefficient of *TOE*, however, loses its significance in a specification with additional control variables reported in column (2) of Table 4. Deals that embed a collar structure and a termination fee in their negotiations are also more likely to exhibit higher cumulative abnormal volume, by about 7.23% and 5.65%, on average. A collar structure implicitly defines a target price range for the takeover agreement. Moreover, a termination fee makes it more likely that a negotiation will be concluded. Thus, both variables are associated with greater certainty about the magnitude of the target's stock price increase, conditional on an announcement being made. This is consistent with a greater likelihood of informed trading in the presence of greater price certainty. All other variables are statistically insignificant. The adjusted R^2 of the regression, 6%, is reasonable, given the likely idiosyncratic nature of the derived statistic for cumulative abnormal trading volume.

We next examine whether market activity variables have an impact on the pre-announcement cumulative abnormal call volume. We include *TRUNUP*, the pre-announcement cumulative abnormal stock return for the target, *TANNRET*, the target's announcement-day abnormal return, *TTPRET1*, the target's post-announcement cumulative abnormal return, and *ARUNUP*, the abnormal stock return for the acquirer before the announcement day. *MKTVOL* denotes the market volume on the day before the announcement. These results are reported in column (2) of Table 4. The pre-announcement run-up in the target's stock price is strongly positively related to the cumulative abnormal volume, consistent with Acharya and Johnson (2010) and Wang (2013). This finding suggests that abnormal options trading activity on the target firm may be instigated by the target firm's positive stock price momentum. We examine this possibility in Section 5. On the other hand, the target's cumulative abnormal announcement return is negatively associated with the cumulative abnormal trading volume for call options. All other variables are statistically insignificant. The coefficients remain robust for large deals that are cash-financed, that have a collar structure, and that have a termination fee. In this final regression specification, the explanatory power increases to 12%. We repeat the analysis for cumulative abnormal volume in put options. While the results are qualitatively similar, the magnitudes of the coefficients are typically smaller and are not reported here.

The insignificant (negative) relation between abnormal call options volume and the abnormal announcement (post-announcement cumulative) return may appear at odds with the notion that the abnormal options activity is informed. The results are nevertheless consistent with the findings in Jayaraman et al. (2001), Spyrou et al. (2011), Klapper (2013), and Podolski et al. (2013). Cao et al. (2005) rely on signed volume to show that buy minus sell call volume order imbalance in the pre-announcement period positively predicts two-day abnormal announcement returns. In unreported regressions, we use signed options volume data, taken from the International Securities Exchange for a subsample of approximately 400 takeovers from 2006 onwards, and confirm the positive relation between pre-announcement call option order imbalance and announcement returns. As we use unsigned volume, the relation is not as straightforward. If much of the insider information gets incorporated into prices in the pre-announcement period (Meulbroek; 1992), one may expect a negative relation between the volume run-up and the announcement return, as the stock price run-up and the announcement effect are negatively correlated for a given offer premium (-14% in

our sample). A priori, the relation between an unsigned volume metric measured over thirty days and the announcement effect is unclear. Rather, we would expect a positive relation between the magnitude of the abnormal pre-announcement options volume and the total premium paid by the bidder, which corresponds to the sum of the pre-announcement run-up (*TRUNP*), the announcement effect (*TANNRET*), and the post-announcement return (*TTPRET1*). Thus, we replace these three variables with the total premium (*TOT_PREMIUM*) and find in column (3) in Table 4 a significantly positive relation, consistent with the interpretation that the options activity is informed.

In columns (4) and (5), we report the results of a logistic regression in which we replace the dependent variable with an indicator variable that takes on the value one, if a deal has cumulative abnormal call options trading volume at the 5% significance level, and zero otherwise. The evidence corroborates the positive relation between the abnormal options activity in the pre-announcement period and variables that are positively correlated with a higher probability of informed trading.

5 Informed vs. Insider Trading

There is a long-standing debate as to whether the stock price run-up before M&A announcements is rationally anticipated or whether it is the result of private information. The pre-announcement run-up in options volume may be due to rational anticipation of upcoming takeover announcements, arising through rumors about upcoming tender offers, speculation due to industry-specific merger waves, or simply because of the superior ability of certain investors to forecast deal activity. Alternatively, it could arise through trading on private information. In this section, we assess the likelihood of the informed options activity being illegal. To this end, we examine deal-by-deal whether publicly available information may explain the pre-announcement run-up in options volume.

5.1 A Legal Definition of Insider Trading

In the U.S., insider trading is regulated under the Securities Exchange Act of 1934 and the responsibility for enforcement lies with the SEC.¹⁰ Registered insiders are bound by the “classical” theory implicit in the antifraud provisions in the Act, which holds them liable if they have traded based on material non-public information from their company, and if they have violated their fiduciary duty.

¹⁰For details on insider trading regulations, see Bainbridge (2007), Crimmins (2013), and Morrison-Foerster (2013).

Outsiders to the firm may legally infer a material “mosaic” conclusion by piecing together multiple pieces of immaterial non-public information. However, they are bound by the “misappropriation” theory implicit in the Act, which prohibits trading based on information that is misappropriated from a third party to whom the investor owes a fiduciary duty.

The boundaries of illegal insider trading are blurry, at best, and the bar on identifying insider trading in court is high.¹¹ It is difficult to draw a clear and precise distinction between a trade that is speculative, informed and legal, and one that is illegal. Thus, we focus on each deal, narrow down on the possibility that the unusual/abnormal pre-announcement options activity might be explained by publicly available information.

5.2 Speculation

Merger activity is pro-cyclical and arrives in industry-specific waves (Andrade et al.; 2001). The earlier findings for the sources of informed trading, namely that cumulative abnormal volumes are significantly related to the run-up in stock prices, may also suggest that the abnormal options activity may simply be the result of positive price momentum on the target’s stock. Thus, speculation may explain the pre-announcement options activity. Such a selection bias would be consistent with the view that takeovers are anticipated.

We examine this possibility by constructing several control samples, matching them based on either the activity in the underlying stock, or firm characteristics.¹² In other words, for each of the 1,859 takeover deals, at the time of the takeover announcement, we look for a similar firm with traded options, and which resembles the takeover firm either in terms of stock market activity measures for the underlying stock, or firm characteristics. For the market-based control sample, we match firms based on three-month moving averages of a firm’s stock return, stock return volatility using the exponential-weighted moving average model, stock trading volume, percentage bid-ask spread to capture illiquidity, and three-month cumulative stock returns to capture the momentum in a firm’s stock. We match only targets for completed deals based on the month prior to the takeover

¹¹The bar on identifying insider trading was raised in a 2014 decision by the Second Circuit in *United States v. Newman*, ruling that it is necessary to show that (a) the trader knew that the information was confidential and illegal, and (b) that inside information was provided in exchange for a benefit of a pecuniary or similarly valuable nature. However, the Supreme Court rejected this Second Circuit’s requirement in December 2016, in *Salman v. United States*.

¹²We explore different combinations of market variables and/or firm characteristics for the matching procedure, using both raw volume and natural log transformations of options volume. We also conduct tests using takeover propensity score matched samples following the methodology in Roberts and Whited (2012). All robustness tests are quantitatively and qualitatively consistent with the reported evidence, and are therefore not reported.

announcement. We sample with replacement, and use the Mahalanobis distance metric to evaluate the “closeness” of the match. Table A-6 in the Internet Appendix documents the matching quality for both control samples. Most variables are statistically indistinguishable between the treatment and control groups, while those that are resemble each other closely in terms of economic magnitude.

The findings reported in Table 5 suggest that the run-up in options trading volume prior to the announcement is unlikely due to investors speculating in the options market after having observed (perhaps) informed trading activity in the underlying shares. We report results for aggregate options volume, and separately for the aggregate call and put volumes, using the MMV model, in which we control for the lagged values of both the dependent and independent variables. The results in Panel A of Table 5 suggest that about 25% of all deals in the treatment sample have positive abnormal call options trading volume, with a lower frequency (16%) of unusual options activity for put options. The average cumulative abnormal total options and call volume is 10,768 and 11,145 contracts, respectively. Both values are statistically significant at the 1% level.

In Panel B of Table 5, we report the results for the control groups using the first best (PS1) and the two first best (PS2) matches. The frequency of deals with statistically significant cumulative abnormal volume at the 5% significance level is lower than in the treatment sample, ranging between 13% and 14%. For call options, the average cumulative abnormal volume is significantly lower in the two control groups than in the treatment groups, as the average cumulative abnormal call volume is only 1,011 and 764 contracts. Importantly, none of the statistics is statistically significant.

In Panel C of Table 5, we report the average treatment effects obtained from a regression of the cumulative abnormal options volume on an indicator variable that takes on the value one if a target belongs to the treatment group, and zero otherwise. In all the regressions, we control for the matching variables to account for residual differences between the treatment and control groups, year fixed effects, and we cluster standard errors by announcement date to account for the possible clustering of announcement dates. These difference-in-difference tests are akin to controlling for both the effects of market activity in the underlying or firm characteristics, and the effects of time. The difference in the average cumulative abnormal volume across the treatment and the two control groups is 10,356 and 10,748 contracts, respectively, and ranges between 11,516 and 12,043 contracts for the aggregated options volume. For put options, the difference for the two sets of matches tends to be considerably smaller, at 1,687 or 768 contracts, respectively, and is not significant. In contrast,

the difference is statistically significant at the 1% significance level for call options and the aggregated options volume. Our findings suggest that speculation is unlikely to be driving the abnormal options volume in the treatment sample, as similar findings are absent from control samples matched on stock market activity and firm characteristics.

The significant differences in options activity across the two groups are visually depicted in Figure 2, which reports the average abnormal volume and average cumulative abnormal volumes in both the treatment and the control group using the first best match. The average abnormal options volume rises significantly ahead of the takeover announcement for the treatment sample (dashed lines), while it fluctuates randomly for the control groups (solid lines).

We conduct two robustness tests. First, we match target firms based on firm characteristics. We use size, measured by the natural logarithm of total assets; industry takeover activity, captured by an indicator variable that equals one if a takeover attempt occurred in the same 4-digit SIC code in the previous calendar year; the presence of at least one institutional block-holder with a minimum 5% equity stake; firm leverage measured as the ratio of total liabilities divided by total assets; and the natural logarithm of the average stock trading volume in the previous calendar year. Second, we match firms based on size, market-to-book ratios, and momentum, defined as either the 12-month cumulative return over the previous calendar year or the past 3-month cumulative stock return in the month prior to the takeover. For brevity, we report the results in the Internet Appendix in Figure A-5 and in Table A-7. Overall, our findings suggest that the unusual options activity is significantly larger in the sample of firms that were takeover targets than in the sample of firms that were not, but closely resemble the takeover targets based on observable industry and firm characteristics.

5.3 Buying the Rumor

Run-ups in trading activity could reflect the market's correct anticipation of future takeover activity based on information "heard on the street" (e.g., Jarrell and Poulsen; 1989). We test this hypothesis using RavenPack News Analytics, a leading global news database widely used in quantitative and algorithmic trading, which contains detailed real-time information on news and rumors about M&A activity.

We rely on the information category "acquisitions-mergers." There are, in total, 88,103 observations for 6,913 different entities (CUSIP identifiers), coming from news sources classified as full

articles, hot news flashes, news flashes, press releases, and tabular material. The bulk of the information on tender offers comes from news flashes, which make up 60.39% of the sample. The two other important categories are full articles and press releases, representing 17.37% and 19.85% of the information, respectively, while hot news flashes and tabular material contribute only marginally to the structured information.

We flag each deal with an indicator that equals one if there exists any deal rumor during the pre-announcement period, and zero if not. We find a rumor or news story on 5,195 different deal-days, corresponding to 877 unique deals from our sample of 1,859 takeovers announced between 1996 and 2012. Most of the news and rumor information appears on the announcement day itself, as shown in Table 6, which illustrates the total number of observations and unique deals in different sample windows. Rumors or news stories exist in the 30-day pre-announcement period for only 170 firms, which corresponds to approximately 9% of our sample, or 13.72% on a proportional basis, given that 1,239 takeovers were announced between January 2000 and August 2012. Most importantly, out of the 467 takeover announcements with significant informed options trading, we associate only 8.57% (40 deals) with news or rumors captured by RavenPack.

It is possible that rumored firms drive the unusual trading activity we document ahead of the takeover announcements. Thus, we investigate whether there is more abnormal volume in options for those firms that have rumors compared to a control group of targets without rumors. Figures 2c and 2d show the differences in average and average cumulative abnormal option trading volumes for the sub-samples with and without rumors in the 30-day pre-announcement period. These tests are based on a natural log transformation of volume. The two sub-samples are statistically indistinguishable from each other, and the average abnormal trading volume is as unusual in the sub-sample with news “heard on the street,” as in the sample without. The same results hold if we screen the sample for news and rumors in the 90-day pre-announcement period.¹³

We provide a visual overview of the sub-sample overlaps using a Venn diagram embedded in Table 7. The outer square outlines the full sample of 1,859 takeovers, which we label Ω . The informed trading sub-sample is characterized by the rectangle A , which indicates that 467 deals are flagged

¹³Using the sentiment scores associated with RavenPack news for the subsample of 170 deals with news and rumors, we confirm previous findings that sentiment explains abnormal volume (Han; 2008). We note that bullish signals lead to more abnormal trading in both calls and puts, while the strength of our previous findings of unusual options activity ahead of takeover announcements was primarily confined to calls. Thus, rumors and news trigger speculation and more trading activity overall, but not necessarily directional trading on the target.

with statistically significant positive cumulative abnormal trading volume at the 5% significance level, corresponding to 25.12% of the entire sample. In the 30 pre-announcement days, we identify news for 170 deals. The news sample is depicted as the rectangle B , and accounts for 9.14% of the overall sample. Out of the 170 takeovers that are associated with news, 40 overlap with the 467 deals flagged for informed options trading, representing 8.57% of the informed trading sample. The remaining overlaps are described in the following subsections. We also find that tests based on a natural log transformation of volume indicate no statistically significant difference in abnormal trading volume between the sub-samples with and without news. Rumors and news about upcoming merger activities are thus unlikely to explain the full amount of directional trading volume on targets ahead of the announcements we document.

5.4 Legal Insider Trading

Registered corporate insiders have access to privileged information. As a result, they must file with the SEC whenever they trade in their company's securities and their derivatives.¹⁴ Even though it is illegal for insiders to trade the target firm's securities prior to a takeover announcement, there is some evidence of such activity in prior research. This could be explained through limited enforceability of insider trading laws, or a lack of prosecution (Arshadi and Eyssell; 1991; Bris; 2005). For example, in the U.S., Agrawal and Nasser (2012) document that insiders increase their net stock purchases prior to takeover announcements.

As prior research has not reported on transactions by registered insiders in target firms' options, we examine the registered derivatives-trading activity, which is logged in Table 2 of the Thomson Reuters insider filing data feeds.¹⁵ We made an elaborate search in the Thomson Reuters filings of transactions by insiders for our 1,859 target firms. We found not a single record of a transaction, purchase or sale, of a derivative security within the 30-day window preceding the announcement. Nevertheless it is possible that the unusual options volume we document stems from tips originating

¹⁴Corporate insiders are defined broadly as people who have "access to non-public, material, insider information," corporate officers, directors, or large block-holders with a stake of 10% or more in the company. They are required to file SEC Forms 3, 4, and 5, and under certain circumstances, Form 144, whenever they trade or intend to trade in their company's securities.

¹⁵We screen all open market derivative transactions, as well as information on the exercise, award, and expiration of stock options, based on the Form 4 filings, which document a change in an insider's ownership position. We examine options, calls and puts, warrants, employee stock options, and group derivative security types with option-embedded features, such as convertibles. We ignore option expirations and swap transactions, and we drop all records that are flagged with the cleansing codes S or A, indicating inaccuracies in the data that are impossible to validate or are missing.

with senior executives at target companies or from former school ties (Cohen et al.; 2010). Indeed, Ahern (2017) reports that tips that lead to illegal insider trades often originate from corporate executives.

5.5 Leakage

In this subsection, we examine whether the abnormal volume effects observed in the options market could be driven by pre-announcement leakage or informed trading in stock markets. There is evidence to suggest that option volume tends to rise in response to positive stock returns (Roll et al.; 2010; Johnson and So; 2012). We test our conjecture in several ways.

First, in all our abnormal volume tests, we systematically control for both contemporaneous and lagged stock returns of the target companies and the overall market. This does not affect the evidence of abnormal options activity in the pre-announcement period. Second, we conduct an event study for abnormal stock returns, and find that only 7.26% (135 deals) of all 1,859 takeovers in our sample exhibit abnormal stock returns at the 5% statistical significance level. Out of the 467 deals that we associate with informed options trading, only 9.64% (45 deals) exhibit abnormal stock returns, as is illustrated by the overlaps of the informed option trading subsample *A* and the subsample with abnormal stock returns *D* in the Venn diagram illustrated in Table 7.

Third, although we do find that about 18.93% of all deals have abnormal stock volumes at the 5% level in the 30-day pre-announcement period, the expected cumulative abnormal log volume for stocks is 1.64, which is about a fifth of what we find in the options market, i.e., 8.59 in terms of logs.¹⁶ Even though stock volume is not directly comparable to (unadjusted) options volume, the net effect from multiplying the options volume by the hedge ratio, i.e., the delta, and the size of 100 shares specified in the standard options contract, would make this difference even wider. This further shows that the magnitude of pre-announcement abnormal volumes is greater in the options market than in the stock market. Most importantly, out of the 467 takeover announcements with abnormal options trading volume, only 181 deals are associated with abnormal stock trading volume (subsample *C* in the Venn diagram illustrated in Table 7). Overall, the abnormal options trading volume for 262 deals (14.09% of the sample) is unlikely due to activity in the stock market.

¹⁶We find that about 24% of all deals have abnormal stock volumes at the 5% level, if we use a natural log transformation of stock volume, which we compare to 33% of all deals that have cumulative abnormal options volume using a natural log transformation. As the relative difference between deals with abnormal stock and options volume is smaller for results using raw volume, we consider these magnitudes to be on the conservative side.

Fourth, we have examined the entire distribution of the option-to-stock volume ratios (see Figure A-4 in the Internet Appendix). We find a significant increase in the ratios of the call-to-stock and the call-to-put volumes, in particular at the right tail of the distribution, but only a modest increase in the ratio of the put-to-stock volume. Dividing the raw trading volume in stocks by 100 to make it comparable to the volume in the options markets (since each option contract is based on 100 shares), we find that the average (median, 90th percentile) call-to-stock volume ratio increases from 7% to 11% (1% to 4%, 15% to 29%) in the pre-announcement period. Similarly, the call-to-put volume ratio increases from 16.83% to 30.72% at the 90th percentile of the distribution, while the put-to-stock volume increases by a more modest amount from 6% to 8% over the 30-day pre-announcement window.

Finally, we studied the lead-lag relation between options and stock volume in the 30-day pre-announcement window. While options and stock volumes are positively related in contemporaneous regressions, options volume is predictive of stock volume, but stock volume is not predictive of options volume. Thus, if anything, the activity in the stock market responds to the activity in the options market, and not the other way round. These findings are consistent with those of Cao et al. (2005) and Liu et al. (2015), among others, who find that the options market displaces the stock market for information-based trading during the periods immediately preceding takeover announcements, but not in normal times.

5.6 Other Explanations: Deal Predictability, Activist Investors, M&A Advisors

We next assess the ability of traders to predict merger activity using publicly observable information. Consistent with the literature (Palepu; 1986; Ambrose and Megginson; 1992; Billett and Xue; 2007; Cremers et al.; 2009), we find that it is difficult to correctly predict, using publicly available information, whether a company is subject to a future takeover threat.¹⁷ Takeover propensity scores are low, about 4% on average, and the explanatory power of the predictability models is not much higher than 5%. Even if some investors have superior ability to process information (Solomo and Scholtes; 2015), it is much less conceivable that they could correctly predict the exact timing of a deal. Thus, it seems unlikely that the abnormal options volume, which is most pronounced in the ten days preceding the public announcement, is due to investors correctly predicting future announcement dates.

¹⁷We report details on the takeover predictability model in the Internet Appendix.

Another source of information comes from registered active investment advisors, who need to file 13D schedules no later than 10 days following the acquisition of beneficial ownership of more than 5% of any class of publicly traded securities. While RavenPack generally captures news associated with 13D filings, not all filings necessarily trigger news and, hence, may be missing in their database. Thus, we need to examine whether any such filings occurred within the 90-day pre-announcement window, using the comprehensive SEC EDGAR database, which contains beneficial ownership reports.¹⁸ In total, we identify 181 13D filings during the 90-day pre-announcement window, of which 74 schedules are filed within the 30-day pre-announcement window, relating to 161 and 70 unique takeover deals, respectively. Out of the 467 deals identified with informed trading, only 3.64% (17 deals) coincide with the filing of beneficial ownership report in the 30-day pre-announcement window.¹⁹ These results are consistent with Collin-Dufresne et al. (2015), who find little evidence of derivatives trading by activist investors, and no difference in call volumes on days when Schedule 13D filers trade, and on days when they do not trade. Abnormal pre-announcement options activity is thus unlikely driven by information derived from 13D filings.

Some authors, using quarterly data, argue that investment banks with M&A advisory desks take equity or options stakes in the target companies during the window between seven and one quarters before the announcement (Bodnaruk et al.; 2009; Lowry et al.; 2016). However, using more granular high-frequency data on broker-level transactions and connections, Griffin et al. (2012) find that such institutional investors do not engage in trading on inside information during the two to 20 days before takeover announcements.

Overall, our results suggest that public sources of information are unlikely to explain the pervasiveness of abnormal options trading activity identified in the 30-day pre-announcement period for about 25.12% (467 deals) of the 1,859 takeover transactions. More specifically, the 236 deals (13%) that we fail to associate with public information channels are likely to represent a lower bound, as 404 deals (21.73%) have abnormal volume in both the stock and the options market (without being associated with public information), yet the options volume leads stock volume in the pre-announcement period. This finding raises the question whether the unexplained pre-announcement option volumes

¹⁸We are indebted to Jan Schnitzler and Ulf Von-Lilienfeld-Toal for sharing this data with us.

¹⁹If we consider filings within 90 days before the announcement, 43 deals have both abnormal options trading volume and 13D filings. We note that any options activity in response to such a filing during the 90 to 30 days before the announcement would introduce a bias against finding evidence of informed options trading, as this would increase the predicted normal volume during the event window. Thus, we are primarily concerned with filings that occur during the 30-day pre-announcement window.

can be related to identified cases of illegal trading activity.

6 SEC Litigation Reports

In this section, we examine the relation between the abnormal options activity identified in the sample of 1,859 takeover transactions and the civil litigations initiated by the SEC for instances of illegal security trading around M&A announcements. We cross-check each takeover flagged for informed trading with the list of deals that were subject to SEC litigation. In addition, we compare the characteristics of option trades that were investigated to those of the abnormal option volumes that we identified in the 30-day pre-announcement period.

We scan more than 8,000 litigation releases concerning civil lawsuits brought by the SEC.²⁰ We extract all cases from the SEC files that encompass trading in stock options around M&A and takeover announcements (i.e., civil complaints against illegal insider trading in options, or, in both stocks and options). We complement missing information in the civil complaint files with information from the criminal complaint files, accessible through the DoJ's Public Access to Court Electronic Records (PACER). A summary overview of the trades and their characteristics is reported in Table 8, and more detailed case-specific information is available upon request. We find that the characteristics of prosecuted trades closely resemble the characteristics of those options that exhibit unusual option volumes and prices, which we find to be pervasive prior to takeover announcements.

6.1 The Characteristics of Insider Trading

In total, we identify 408 M&A transactions involved in insider trading litigations between January 1990 and December 2013; among these M&A transactions, 258 are investigated due to insider trading in stocks only, and 150 involve insider trading in options. About 31% of these cases (47 deals) cite insider trading in options only, while the remaining 103 cases involve illicit trading in both options and stocks. The large number of investigations for stock trades relative to those for option trades stands in contrast to our finding of pervasive abnormal call option trading volumes that are greater than abnormal stock volumes.

Out of the 150 SEC cases with illegal option trades, 131 correspond to our sample period, which

²⁰The litigation reports are publicly available on the SEC's website, <https://www.sec.gov/litigation/litreleases.shtml>.

stretches from January 1996 to December 2012. Several of the litigated cases do not appear in our sample, one reason being the sample selection criteria for our study. On the other hand, some prominent cases of insider trading, such as the 2004 JPM Chase-Bank One merger, do not appear in the SEC database. We have three potential explanations for these discrepancies.

First, the SEC only reports civil litigations. If a case is deemed criminal, then the DoJ will handle it and it will usually not appear in the SEC records. We believe this reason to be an unlikely cause, given that a case typically does not come under criminal investigation without first being investigated by the SEC. Our interpretation is based on several discussions with securities law firms and prosecutors. It is also corroborated by Ahern (2017), who identifies only two cases among all (not just for M&As) the DoJ criminal complaints for insider trading that do not appear among the SEC civil litigation records. Second, the SEC may refrain from divulging the details of a case to protect the identity of a whistleblower. In these instances, if the case is settled out of court, it will not appear in the public record. Third, the SEC is unlikely to litigate if there is little chance of indictment. Despite these biases, 90 of the SEC litigation cases with illegal option trades are covered by our study, and 154, if we include those litigations of takeovers that feature only illegal stock trades. Assuming that the publicly disclosed deals represent all litigated cases, the public records suggest that the SEC engaged in litigation in about 8.28% of the 1,859 takeover deals in our sample (see the informed trading subsample *E* in Table A-9). The four-dimensional Venn diagram on the top right figure of Table A-9 in the Appendix illustrates that, out of the 467 deals with informed options trading activity, 43 deals (9.21%) are litigated by the SEC/DoJ. Note also that SEC/DOJ initiated a litigation for only 10.21% (24 deals) of the 236 transactions with informed options activity that is unlikely explained by public sources of information.

The statistics in Table 8 indicate that, for the takeovers available in the SEC litigation reports, 59.33% of all cases are cash-financed transactions. Unreported statistics suggest that only 23.33% are purely stock-financed, while hybrid financing structures account for 12% of the litigation sample, and the financing structure is unknown for the remaining 5.33% of the sample. Investors with private information are less likely to trade on stock-financed announcements, as the announcement return is typically higher for cash deals. This is consistent with our finding of a greater cumulative abnormal call option volume for such transactions. The average profit reaped through “rogue trades” (in both options and stocks) during our sample period is 1.084 million USD. Meulbroek (1992) and Frino

et al. (2013) report median insider profits of 24,673 USD and 26,860 USD, respectively, for illegal trading in stocks. For trades on the target firms, this profit arises from transactions that are almost exclusively purchases of OTM call options, at single or multiple strike prices.

The litigation reports reference put trades in only 6% of all cases. For 22 of these 25 put trades, we can identify the trading direction, which suggests that these were all sales of put options, consistent with the hypothesis that insiders would buy OTM calls and/or sell ITM puts. Table 8 shows that the average ratio of the stock price to the strike price, in the case of call options purchased on the target, is 93%. Only 25 observations ($\approx 6\%$) have a moneyness ratio above 1.05, the cut-off level we defined for ATM options. Out of 25 put option trades on the targets, on the other hand, the average ratio of the stock price to the strike price is 97.29%, which is within our definition of ATM, but 12 of all these trades relate to sales of ITM put options with an average ratio of the stock price to the strike price above 105%.

Furthermore, the insider trades are primarily executed in short-dated options, with an average time to expiration of 1.87 months. We note that there is a large variation in the timing of trades, the average inside trader transacting 21 days before the announcement date. However, the median trade occurs 11 days prior to the announcement. In contrast, Frino et al. (2013) document that insiders trade in target firms' stocks on average 7 days before the announcement. This finding is supportive of the results discussed in subsection 5.5, which suggest that options volume leads stock volume in the pre-announcement period.

It takes the SEC, on average, 644 days to publicly announce its first litigation action in a given case. Thus, assuming that the litigation releases appear shortly after the actual initiation of investigations, it takes the SEC almost two years, on average, to prosecute a rogue trade. The fines, including disgorged trading profits, prejudgment interest and civil penalty, if any, appear large enough to adequately recuperate illicit trading profits. The average fine is, at 1.889 million USD, about double the average rogue profit. This is, however, largely driven by cases related to 2007, which exhibit a ratio of the average fine to the average profit of about 4.39. Fines are also larger than those reported earlier for illegal trading in stocks. The median penalty reported by Meulbroek (1992) and Frino et al. (2013) is 21,000 USD and 67,511 USD, respectively. Finally, the typical insider trade involves more than one person, and is often a network, as documented in detail by Ahern (2017). We find that the average number of defendants is four.

To summarize, the bulk of the prosecuted trades relate to purchases of plain vanilla, short-dated OTM call options written on target companies. The litigated trades are, on average, approximately 7% OTM, executed within the 21-day period before the announcement, and most frequently related to cash-financed deals. There is some indicative evidence of sales of ITM put options on the target companies. The characteristics of allegedly illegal trades closely resemble those of the abnormal options volumes of the 467 deals that we flag for informed trading among the 1,859 takeover transactions. These deals exhibit abnormal option trading volumes that are particularly pronounced for OTM and short-dated call options.

6.2 The Determinants of Insider Trading Litigation

We next examine the determinants of insider trading litigation (i.e., what characteristics attract SEC action). Unfortunately, we are unable to distinguish whether certain characteristics reflect deals that are more prone to insider trading, or whether certain company or market attributes more easily attract the attention of the SEC. For example, the SEC may be more attentive during specific market conditions and/or to a certain type of company.²¹

We examine the impact of takeover deal characteristics on the likelihood of SEC investigation using a logit model. The indicator variable *SEC* takes the value one, if a deal has been litigated by the SEC, and is zero otherwise. We estimate the following logit model as a benchmark:

$$\begin{aligned} Pr(SEC = 1) = F(\beta_0 + \beta_1 SIZE + \beta_2 CASH + \beta_3 CHALLENGE + \beta_4 COMPLETE \\ + \beta_5 TOE + \beta_6 PRIVATE + \beta_7 COLLAR + \beta_8 TERM + \beta_9 FRIENDLY + \beta_{10} US + \gamma_t), \end{aligned} \quad (2)$$

where $F(\cdot)$ defines the cumulative distribution of the logistic function, and all explanatory variables are categorical variables that take the value one if a condition is met, and zero otherwise. *SIZE* takes the value one if the transaction is larger than the median deal value. *CASH* denotes cash-financed takeovers, *CHALLENGE* denotes deals that have been challenged by a second bidder, *COMPLETE* denotes completed deals that were not withdrawn and did not fail, and *TOE* denotes whether a bidder already has a toehold in the target firm. *PRIVATE* equals one, if the acquirer privatized the target post-acquisition. *COLLAR* denotes transactions with a collar structure. *TERM* is one for deals that have a termination fee if takeover negotiations fail. *FRIENDLY* denotes the

²¹Our discussions with the regulator suggest that the resource-constrained SEC is more likely to litigate cases that have a greater chance of resulting in a conviction and that have generated substantial illicit trading profits.

deal attitude. *US* is one, if the bidder is a U.S.-based company. All specifications contain year fixed effects.

We report the logit coefficients (and odds ratios in parentheses), using Firth's method for bias reduction in logistic regressions, in Table 9. The evidence in column (1) suggests that the likelihood of SEC litigation is higher for larger and completed deals that are initiated by foreign bidders. A transaction with a deal value greater than the median takeover deal value is 1.87 times more likely to face litigation. This evidence is consistent with Podolski et al. (2013) and Wang (2013), who find size to be an important predictor of SEC litigation. The log-odds ratio suggests that an acquisition undertaken by a foreign bidder is roughly twice as likely to be prosecuted as a takeover initiated by a U.S.-based bidder. Completed deals are positive predictors of options litigation, as a withdrawn or rumored deal is almost three times less likely to be investigated. The pseudo- R^2 of the regression is reasonable, with a value of 11%. We further test for the importance of the offer premium (*PREM1D*) and the offer price (*PRICE*) in affecting the probability of litigation. The results in column (2) indicate that both the offer premium and the offer price are positively related to the probability of SEC litigation, although the magnitudes of the odds ratios are just above one.

Next, we test whether we can predict the SEC litigations based on the stock price behavior of the parties involved in the transaction. We estimate an augmented logit model and include *TRUNUP*, the target's pre-announcement cumulative abnormal stock return, *TANNRET*, the target's announcement-day abnormal return, *TTPRET1*, the target's post-announcement cumulative abnormal return, and *ARUNUP*, the acquirer's abnormal stock return before the announcement day, and report the results in column (3) of Table 9. Only the target's post-announcement cumulative abnormal return is highly statistically significant. The coefficient of 2.10 suggests that a target with a 1% higher cumulative abnormal post-announcement return is approximately eight times more likely to be investigated. This corresponds to a marginal effect of 4.8%, keeping all other variables at their median levels.

We also check whether the market environment in the pre-announcement period has predictive ability for the SEC litigations. Thus, we augment the base model with *MKTVOL*, the market volume on the day before the announcement, and *ABNORMVOLC*, the target's total abnormal call trading volume during the 30-day pre-announcement period, and report the results in column (4). None of these variables exhibits statistical significance in explaining the SEC civil litigations. This

is surprising, as we do find that 25% of all deals in our sample feature an abnormal options volume that is difficult to associate with public sources of information, and many of these are not litigated.²² In contrast to our results, Wang (2013) reports a positive relation between pre-announcement call options volume and the probability of SEC litigation, without controlling for the influence of common variation. In unreported results, we similarly find a significant positive relation if we omit the year fixed effects. This evidence supports the conjecture that there may be time variation in regulatory enforcement.

DeMarzo et al. (1998) suggest that it may be optimal to prosecute traders with inside information only after large price moves or after large volume transactions, and not to penalize their small trades. Thus, it could be cost-effective and more impactful for a resource-constrained SEC to pursue illicit transactions relating to the securities of larger-sized firms. On the one hand, our results provide support for DeMarzo et al. (1998), given that SEC litigation is more likely for deals with large transaction values, higher bid prices, and higher offer premiums. It is difficult, however, to differentiate whether traders with inside information prefer to trade ahead of transactions involving larger companies, which typically have more liquid options markets and provide for greater opportunity to hide, or whether the SEC is more likely to go after large-scale deals that are easier to detect and more broadly covered in the financial media. On the other hand, we do not find evidence that the odds of regulatory action are higher after observing large abnormal options volumes. Finally, we find that the odds of litigation are higher for deals that are initiated by foreign acquirers. This may indicate that rogue traders focus on foreign jurisdictions in order to gain and exploit their private information. Overall, the number of civil litigations initiated by the SEC because of illicit option trading ahead of takeovers appears modest in light of the pervasiveness of abnormal options trading that we document for 25% of the takeover transactions in our sample.

7 Conclusion

We quantify the prevalence of informed trading activity in options of target firms ahead of unexpected takeover announcements in the U.S. from 1996 until 2012. We find that 25% of all deals in our sample exhibit statistically significant abnormal trading volumes in options, and that this activity appears

²²In unreported tests, we examine whether there is any fundamental difference between those SEC cases that were pursued because of alleged insider trading in options and those that were investigated because of allegedly illicit trading in stocks. Our previous conclusions remain largely unchanged.

to be informed. Statistical tests suggest that the odds of the abnormal volume arising out of chance are, at best, one in a million.

Our key contribution is the deal-by-deal examination of the sources of informed trading to assess the likelihood of the informed options activity being illegal. We show that the magnitude of this abnormal activity is unlikely explained by speculation, news and rumors, trading by registered insiders, stock market leakage, or investors' ability to predict or time the takeover announcement. Cross-checking the overlaps between all these plausible explanations, we find that the activity for a minimum of 13% of all deals in our sample is difficult to associate with public sources of information.

The prevalence of informed trading appears more pervasive than would be expected based on the number of prosecutions, as the SEC litigates illegal stock and option trades for only 8.28% of takeovers, and for only 43 out of the 467 of the deals in our sample that display significant informed options activity. Yet, the characteristics of SEC-litigated insider trades in options ahead of takeover announcements closely resemble the characteristics of the (mostly unprosecuted) pre-announcement options activity that we flag as informed.

Guercio et al. (2017) argue that illegal insider trading in the stock market has decreased in response to more aggressive enforcement. Our work, in contrast, suggests that it could have moved to the options market. This is thought-provoking, especially if there appears to be substantial insider trading in many countries with less sophisticated markets than the U.S. (Griffin et al.; 2011), which is the focus in our study. The characterization of how and where informed investors trade is not only of interest to economists studying the information structure of asset markets around takeover announcements, but also to regulators and policy makers. Our findings call for further regulatory examination of abnormal option volumes ahead of takeover announcements.

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Table 1: Informed Trading Ahead of Takeover Announcements. In this table, we summarize the literature studying informed options activity in the pre-announcement period of mergers and takeovers, focusing on target options. We describe each paper's geographical focus (Country), whether it examines either stock or options activity, or both, whether it examines both target and acquirer options, and whether it quantifies the pervasiveness of informed options trading (Quantification). + indicates that the authors do not provide statistical significance for cases with increased options activity. For the data characteristics, we report the number of target deals, (* refers to acquirer deals), the subsample of deals with option data availability, and the study's time period. We indicate the sources of informed trading activity studied for abnormal activity in target options: rumors/news, speculation, options trading by registered insiders, stock market leakage, merger predictability, trading by activist investors, and trading by connected M&A advisory desks. We also indicate whether the study examines the occurrence of illegal insider trading litigations initiated by the SEC, and whether the study describes the nature and characteristics of option trades prosecuted by the SEC.

Study	Paper's Focus		Data Characteristics		Sources of Informed Trading on Targets										
	Country	Stocks	Options	Quantification	Acquirer Target	# Target Deals	Time Period	Rumors/News	Speculation	Registered Insiders	Stock Market Leakage	Activist Investors	M&A Advisors	SEC Litigation . . .	... Characteristics
Authors and Year	Jayaraman et al. (1991)	U.S.	✓	✓ ⁺	✓	27	75-84								
	Levy and Yoder (1993)	U.S.	✓	✓	✓	21	82-85								
	Jayaraman et al. (2001)	U.S.	✓	✓ ⁺	✓	33	86-96								✓ [*]
	Cao et al. (2005)	U.S.	✓	✓	✓	78	86-94								
	Arnold et al. (2006)	U.S.	✓	✓	✓	401	94-00								
	Acharya & Johnson (2010)	U.S.	✓	✓	✓	212	84 00-06								
	Spyrou et al. (2011)	U.K.	✓	✓	✓	3,875	87-06								
	Clements & Singh (2011)	U.S.	✓	✓	✓	393	01-06	✓							
	Shafer (2012)	U.S.	✓	✓	✓	824	96-10			✓					
	Klapper (2013)	U.S.	✓	✓	✓	2,390	96-12	✓							
	Wang (2013)	U.S.	✓	✓	✓	3,124	96-08	✓							
	Podolski et al. (2013)	U.S.	✓	✓	✓	815	96-08							✓	
	Ordu & Schweizer (2015)	U.S.	✓	✓	✓	5,986*	96-08								✓
	Liu et al. (2015)	U.S.	✓	✓	✓	3,711	96-11				✓				
	Chan et al. (2015)	U.S.	✓	✓	✓	2,372	96-10								
	Huang and Tung (2016)	U.S.	✓	✓	✓	291	96-13								
	Lowry et al. (2017)	U.S.	✓	✓	✓	703	00-11						✓		
The present study (2017)	U.S.	✓	✓	✓	1,859	96-12	✓	✓	✓	✓	✓	✓	✓	✓	

Table 2: Descriptive Overview of Takeover Sample

In Panel A, we describe the deal characteristics of the 1,859 takeovers between January 1996 through December 31, 2012. We report the number of deals (No.) and the corresponding sample proportions (% of Tot.). In addition, we report how many of the deals are classified as completed, friendly, hostile, involving a target and acquirer in the same industry, challenged, or having a competing bidder, a collar structure, a termination fee or a bidder with a toehold in the target company. All characteristics are reported for the overall sample (column *Total*), as well as for different offer structures: cash-financed (*Cash Only*), stock-financed (*Shares*), a combination of cash and stock financing (*Hybrid*), other financing structures (*Other*), and unknown (*Unknown*). In Panel B, we illustrate the financial statistics of the deals. We report the transaction value (*DVal*) in million USD and the offer premium, defined as the ratio of the offer price to the target's closing stock price. P1d (P1w, P4w) refers to the premium, one day (one week, four weeks) prior to the announcement date, in percentage terms.

Panel A: Deal Information													
Offer Structure		Cash Only		Hybrid		Other		Shares		Unknown		Total	
Description	No.	% of Tot.	No.	% of Tot.	No.	% of Tot.	No.	% of Tot.	No.	% of Tot.	No.	% of Tot.	% of Tot.
No. of Deals	903	48.6%	415	22.3%	80	4.3%	403	21.7%	58	3.1%	1,859	100.0%	
Completed Deals	746	40.1%	357	19.2%	67	3.6%	339	18.2%	33	1.8%	1,542	82.9%	
Friendly Deals	805	43.3%	379	20.4%	69	3.7%	382	20.5%	42	2.3%	1,677	90.2%	
Hostile Deals	35	1.9%	14	0.8%	3	0.2%	7	0.4%	4	0.2%	63	3.4%	
Same-Industry Deals	379	42.0%	280	67.5%	39	48.8%	268	66.5%	27	46.6%	993	53.4%	
Challenged Deals	111	6.0%	55	3.0%	7	0.4%	32	1.7%	11	0.6%	216	11.6%	
Competing Bidder	83	4.5%	32	1.7%	3	0.2%	20	1.1%	4	0.2%	142	7.6%	
Collar Deal	4	0.2%	54	2.9%	3	0.2%	52	2.8%	7	0.4%	120	6.5%	
Termination Fee	698	37.5%	352	18.9%	51	2.7%	292	15.7%	29	1.6%	1,422	76.5%	
Bidder has a Toehold	42	2.3%	11	0.6%	2	0.1%	7	0.4%	3	0.2%	65	3.5%	

Panel B: Deal Financials													
Offer Structure		Cash Only		Hybrid		Other		Shares		Unknown		Total	
Description	Mean	Sd	Mean	Sd	Mean	Sd	Mean	Sd	Mean	Sd	Mean	Sd	
DVal (mil)	\$2,242.0	\$4,147.2	\$5,880.9	\$10,071.5	\$5,074.2	\$10,387.7	\$5,429.8	\$15,158.5	\$1,635.7	\$2,503.7	\$3,848.4	\$9,401.3	
P1d	33.6%	31.7%	28.5%	27.5%	25.1%	40.5%	28.3%	39.5%	33.3%	29.6%	31.0%	33.1%	
P1w	36.6%	31.0%	32.4%	29.1%	29.5%	42.5%	33.6%	61.5%	33.4%	29.8%	34.7%	39.8%	
P4w	41.1%	35.6%	35.0%	32.4%	31.2%	46.1%	36.7%	45.3%	38.0%	33.6%	38.3%	37.7%	

Table 3: Positive Abnormal Options Trading Volume on Target Companies

Panel A reports the number (#) and frequency (freq.) of deals with statistically significant positive cumulative abnormal volume at the 5% significance level for the target companies, as well as the average cumulative abnormal volume ($E[CAV]$) and corresponding t -statistic (t_{CAV}), computed using heteroscedasticity-robust standard errors. All results are reported separately for call options, put options, and for the aggregate option volume. Results stratified by moneyness are based only on those options expiring after the announcement date. The estimation window starts 90 days before the announcement date and runs until 30 days before the announcement date. The event window stretches from 30 days before until one day before the announcement date. In Panel B, we report the results of t -tests for the differences in the average cumulative abnormal volumes across moneyness categories: out-of-the-money (OTM), in-the-money (ITM), and at-the-money (ATM). We report the difference in average cumulative abnormal volume (Diff), the standard error (s.e.) and the p -value (p -val).

Panel A:	Magnitude and Frequency of Cum. Abnormal Volume Deals					
	All	Calls	Puts	All	Calls	Puts
	All Options - Target			OTM Options - Target		
Sign. t -stat 5% (#)	446	467	304	423	408	451
Sign. t -stat 5% (freq.)	0.24	0.25	0.16	0.23	0.22	0.24
$E[CAV]$	10,385	8,946	1,559	5,071	3,380	1,417
t_{CAV}	3.76	5.77	1.04	5.44	5.46	3.34
	ATM Options - Target			ITM Options - Target		
Sign. t -stat 5% (#)	341	343	362	393	482	396
Sign. t -stat 5% (freq.)	0.18	0.18	0.19	0.21	0.26	0.21
$E[CAV]$	1,652	1,156	457	2,526	1,540	984
t_{CAV}	2.84	2.65	1.98	4.71	6.40	2.42
Panel B:	Differences in Cum. Abnormal Volume across Moneyness					
	Diff	s.e.	p -val	Diff	s.e.	p -val
	All Options - Target			Call Options - Target		
OTM-ATM	3,419	722	0.00	2,224	531	0.00
OTM-ITM	2,544	669	0.00	1,840	561	0.00
ATM-ITM	-874	632	0.17	-384	444	0.39
	Put Options - Target					
OTM-ATM	960	450	0.03	—	—	—
OTM-ITM	433	367	0.24	—	—	—
ATM-ITM	-527	429	0.22	—	—	—

Table 4: Characteristics of Cumulative Abnormal Call Volume

In columns (1) and (2), we report generalized least squares (GLS) regression results from the projection of cumulative abnormal call option log-volume ($CABVOL_C$, standardized by the average predicted volume during the event window) on a set of takeover characteristics and market activity measures. Columns (3) and (4) report logit coefficients (odds ratios in parentheses) from logistic regression results where the dependent variable takes on the value one if the cumulative abnormal call options trading volume during the 30 pre-announcement days is statistically significant at the 5% level, and zero otherwise. $SIZE$ quantifies the takeover deal value. $CASH$ is a categorical value taking the value one if the deal is a cash-financed takeover and zero otherwise, TOE has the value one if a bidder already has a toehold in the target company, $PRIVATE$ equals one if the acquirer privatizes the target post-acquisition, $COLLAR$ takes the value one for transactions with a collar structure, $TERM$ is one for deals that have a termination fee that applies if the takeover negotiations fail, $FRIENDLY$ has the value one if the deal attitude is considered to be friendly, and US is one if the bidder is a U.S.-based company, and zero otherwise. $TRUNUP$ denotes the pre-announcement cumulative abnormal stock return for the target, $TANNRET$ denotes the target's announcement abnormal return, $TTPRET1$ refers to the target's post-announcement cumulative abnormal return, and $ARUNUP$ is the abnormal stock return for the acquirer before the announcement day. $MKTVOL$ is the market volume on the day before the announcement. Each regression contains year fixed effects ($YEAR FE$), and standard errors are clustered by announcement day. We report the number of observations (Observations) and the adjusted R^2 . ***, ** and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

VARIABLES	(1) $CABVOL_C$	(2) $CABVOL_C$	(3) $CABVOL_C$	(4) $I(CABVOL_C)$	(5) $I(CABVOL_C)$
SIZE	3.32** (1.34)	2.44* (1.29)	3.31** (1.33)	0.23** (1.26)	0.21* (1.24)
CASH	6.37*** (1.53)	5.49*** (1.54)	5.15*** (1.59)	0.42*** (1.52)	0.43*** (1.53)
TOE	-5.58* (2.94)	-3.38 (2.71)	-4.82* (2.92)	-0.33 (0.72)	-0.27 (0.76)
PRIVATE	0.12 (1.97)	0.06 (1.91)	0.88 (2.00)	-0.09 (0.91)	-0.13 (0.88)
COLLAR	7.23** (2.94)	6.47** (2.85)	6.52** (2.95)	0.41* (1.50)	0.43** (1.54)
TERM	5.65*** (1.83)	4.57** (1.80)	4.93*** (1.83)	0.20 (1.22)	0.21 (1.23)
FRIENDLY	3.04 (2.36)	1.91 (2.30)	2.85 (2.37)	0.14 (1.15)	0.06 (1.06)
US	-2.45 (1.91)	-1.71 (1.88)	-2.54 (1.92)	-0.35** (0.70)	-0.31** (0.74)
TRUNUP		24.30*** (2.88)			1.01*** (2.75)
TANNRET		0.57 (4.56)			0.74 (2.10)
TTPRET1		-7.84* (4.08)			-1.15** (0.32)
ARUNUP		-4.52 (4.27)	4.21 (4.36)		-0.97** (0.38)
MKTVOL		-3.85** (1.95)	-1.91 (2.02)		-0.06 (0.94)
TOT_PREMIUM			4.24** (1.75)		
Constant	-1.37 (2.79)	15.25* (8.66)	5.68 (9.01)	-0.82** (0.44)	-0.62 (0.54)
Observations	1,859	1,859	1,859	1,859	1,859
adj.R ² / ps.R ²	0.056	0.123	0.060	0.035	0.054
YEAR FE	YES	YES	YES	YES	YES

Table 5: Positive Abnormal Trading Volume in Treatment and Control Groups

In this table, we report the number (#) and frequency (freq.) of deals with statistically significant positive cumulative abnormal volume at the 5% significance level, as well as the average cumulative abnormal volume ($E[CAV]$) and corresponding t -statistic (t_{CAV}), computed using heteroscedasticity-robust standard errors. In Panel A, we report results for the treatment group, and Panel B for the matched control groups using the first (and second) best matches. Targets are matched based on the return, bid-ask spread, stock volume, and stock volatility, using 3-month moving average values from the month prior to the announcement. Panel C reports the average treatment effects obtained from a regression of the cumulative abnormal options volume on an indicator variable that takes on the value one if a target belongs to the treatment group, and zero otherwise. In all regressions, we control for the matching variables, year fixed effects, and standard errors are clustered by announcement date to account for cross-sectional correlation due to possible clustering of announcement dates.

Panel A:		Magnitude and Frequency of Cum. Abnormal Volume Deals Treatment Group PS0 - Takeover Sample				
	All		Call		Put	
Sign.t-stat 5% (#)	296		321		220	
Sign.t-stat 5% (freq.)	0.22		0.24		0.16	
$4E[CAV]$	10,768		11,145		746	
t_{CAV}	2.74		4.94		-0.12	
Panel B:		Magnitude and Frequency of Cum. Abnormal Volume Deals Control Group PS1 Best Match			Control Group PS2 Two Best Matches	
	All	Call	Put	All	Call	Put
Sign.t-stat 5% (#)	161	158	167	326	329	319
Sign.t-stat 5% (freq.)	0.13	0.13	0.14	0.14	0.14	0.13
$E[CAV]$	872	1,011	467	990	764	419
t_{CAV}	-0.01	0.54	-0.47	0.07	0.43	-0.33
Panel C:		Differences in Cum. Abn. Volume Deals across Treat. & Control Groups				
Treatment1	12,043*** (3,471)	10,356*** (2,751)	1,687 (2,108)			
Treatment2				11,516*** (3,251)	10,748*** (2,625)	768 (2,405)
CONTROLS	YES	YES	YES	YES	YES	YES
YEAR FE	YES	YES	YES	YES	YES	YES
CLUSTER	YES	YES	YES	YES	YES	YES

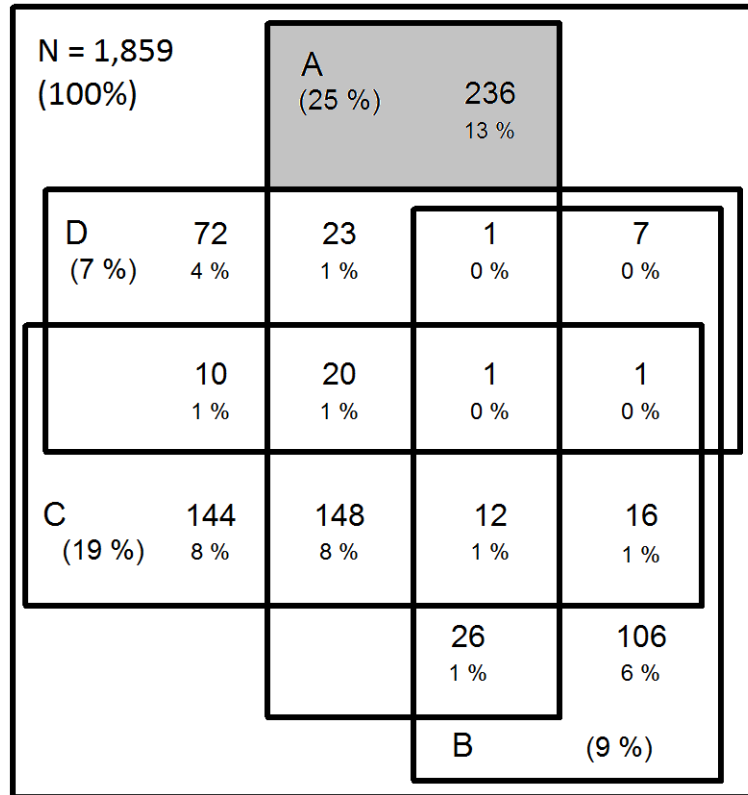
Table 6: Frequency of News around Takeover Announcements

In Table 6, we report the frequency of news around takeover announcements. The information is from RavenPack News Analytics, which extracts textual information from major publishers, such as *Dow Jones Newswires*, *the Wall Street Journal*, *Barron's*, regulatory and public relation feeds and over 19,000 other traditional and social media sites. The company transforms the information into a structured data feed that can be used in quantitative analysis. Using the data from January 2000 to August 2012, we report the number of news and rumor items (*Obs*) and the corresponding number of takeover deals (*Deals*) recorded during different time windows around the announcement day. The window titled “0” relates to news and rumors on the announcement day. All other columns refer to different time windows in the pre- or post-announcement period.

Time Window	[-150,-91]	[-90,-31]	[-30,-21]	[-20,-11]	[-10,-6]	[-5,-1]	[0]	[1,5]	[5,10]	[-150,-91]	[-30,-1]	[-90,-1]
Obs	258	290	67	67	31	74	660	252	140	1839	239	529
Deals	188	204	55	59	29	63	659	218	113	407	170	299

Table 7: Sources of Informed Options Trading

In this table, we document the sources of informed options trading for a sample of 1,859 takeovers from January 1996 to December 2012. Of the 1,859 takeovers, 467 (25.12, subset A) have abnormal options trading volume at the 5% significance level over the 30-day pre-announcement period. The sample of news and rumors amounts to 170 deals (9.14%, subset B); 352 of all deals (18.93%, subset C) have abnormal stock trading volume in the run-up to the announcement, while 135 deals (7.26%, subset D) have abnormal stock returns in the run-up to the announcement. In total, 427 of all deals (22.97%) have abnormal options trading volume ahead of the announcement, as well as abnormal returns and volume in the underlying stock, without any news or rumors; 236 of all deals (13%, grey area) have only abnormal options trading volume and are difficult to associate with public sources of information.



Set	#	%	Cum%	A	B	C	D	Description
Ω	1,859	100.00	100.00	✓	✓	✓	✓	Takeover Sample
A	467	25.12	25.12	✓	✗	✗	✗	Abnormal Options Volume (A)
$\Omega \setminus A$	1,392	74.87	100.00	✗	✓	✓	✓	No Abnormal Options Volume
B	170	9.14	9.14	✗	✓	✗	✗	News & Rumors (B)
$\Omega \setminus B$	1,689	90.86	100.00	✓	✗	✓	✓	No News & Rumors
C	352	18.93	18.93	✗	✗	✓	✗	Abnormal Stock Volume (C)
$\Omega \setminus C$	1,507	81.07	100.00	✓	✓	✗	✓	No Abnormal Stock Volume
D	135	7.26	7.26	✗	✗	✗	✓	Abnormal Stock Returns (D)
$\Omega \setminus D$	1,724	92.74	100.00	✓	✓	✓	✗	No Abnormal Stock Returns
$(A \cap C \cap D) \setminus B$	427	22.97	22.97	✓	✗	✓	✓	A, C, D, No B
$\Omega \setminus (A \cap B \cap C \cap D)$	1,036	55.73	55.73	✗	✗	✗	✗	No A, B, C, D
$A \setminus (B \cup C \cup D)$	236	12.69	68.42	✓	✗	✗	✗	A, No B, C, D

Table 8: SEC Litigation Reports

In Table 8, we provide summary statistics on a sub-sample of litigation releases concerning civil lawsuits brought by the SEC. The column *SEC LRs* indicates the number of M&A deals in each calendar year (*Year*) that have been subject to litigation by the SEC. The column *Cash* indicates the number of litigated deals that are cash-financed (if the information is available). The column *ABS Sample* refers to our sample of takeover deals. The column *Illicit Profits* is the average number of illicit profits reaped in the litigated cases (from stocks and options) and the column *Fines* reports the average yearly fine imposed in the litigations (total amount including disgorged trading profits, prejudgment interest and civil penalty, if any). The column *Days to Lit.* denotes the average number of days between the announcement date and the first filed litigation report. The column *Moneyess S/K* provides information about the average moneyness of the prosecuted call option trades on the target. The column *Option Mat.* presents the average time to maturity (in months) of all options trades under litigation, and the column *Days to Ann.* reports the average number of days between the first unusual option trade and the announcement date. The last column, *Defend.*, shows the yearly average number of defendants. A ⁺ in the first column indicates years with M&As involved in a litigation for alleged illegal trading on the acquiring company. In total, there are only five cases involving the acquirer in a deal. The last two rows report the sample averages over the entire period for which we have information on SEC litigations, as well as over the shorter sample period, 1996 to 2012, that we cover in our analysis of unusual options trading.

Year	SEC LRs	Cash	ABS Sample	Illicit Profits	Fines	Days to Lit.	Moneyess S/K	Option Mat.	Days to Ann.	Defend.
1990	1	0	.	650,000	.	.	.	.	.	17
1993	2	0	.	87,000	72,171	1,514	.	.	13	6
1994	5	2	.	156,690	281,480	883	0.88	1.00	4	3
1995 ⁺	4	2	.	400,319	650,060	1,026	0.93	2.78	26	14
1996	4	2	70	377,612	903,343	456	0.93	0.50	3	2
1997 ⁺	4	1	133	480,367	1,471,178	350	1.02	1.50	2	3
1998	8	3	175	1,443,723	648,023	369	0.89	1.20	8	8
1999	2	2	217	295,676	57,880	1,088	0.94	1.00	4	14
2000	8	4	164	221,340	192,995	915	1.09	1.00	4	2
2001	3	1	86	232,533	270,662	1,212	0.95	0.25	0	4
2002	1	0	36	250,000	61,714	933	.	.	72	4
2003	3	0	54	372,404	905,647	689	0.94	1.85	12	3
2004	2	2	72	1,242,665	1,743,741	438	0.98	1.38	3	.
2005	11	5	109	879,829	1,499,516	841	0.97	1.17	14	6
2006 ⁺	14	10	119	1,001,278	637,230	552	0.97	0.93	13	4
2007	24	17	159	1,396,619	6,132,891	874	0.90	1.83	25	3
2008	6	5	98	1,226,436	1,243,423	719	0.93	3.12	18	2
2009	6	3	74	2,684,571	827,898	634	0.89	2.63	43	2
2010 ⁺	20	15	93	604,633	1,993,138	489	0.94	1.33	17	2
2011	8	4	114	914,337	263,969	832	0.83	3.83	44	4
2012	7	6	86	2,762,365	352,327	211	0.93	1.11	15	7
2013	7	5	.	1,493,579	5,069,953	144	0.96	1.56	8	2
90-13	7	9	.	1,037,617	1,888,521	644	0.93	1.88	21	4
96-12	8	10	109	1,084,162	1,937,595	635	0.93	1.87	21	4

Table 9: SEC Predictability Regressions

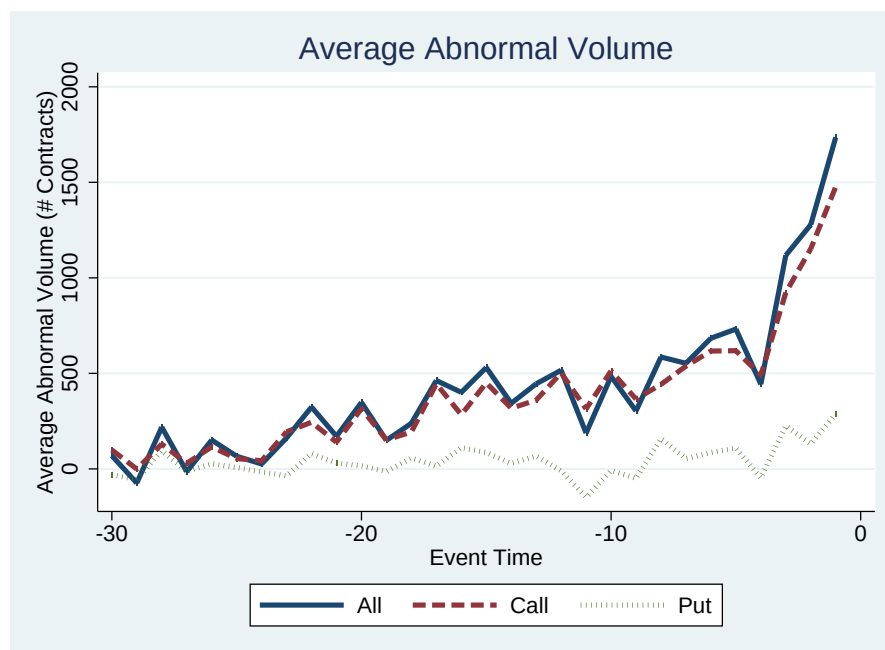
In Table 9, we report logit coefficients from the logistic regressions (odds ratios in parentheses). The dependent variable *SEC* takes the value one if there was litigation in respect of a deal involving options, and zero otherwise. The explanatory variables take the value one if a condition is met, and zero otherwise: *SIZE* takes the value one for deals with a value greater than the median takeover deal value, *CASH* for cash-financed takeovers, *CHALLENGE* for challenged deals, *COMPLETE* for completed transactions, *TOE* if a bidder already has a toehold in the target company, *PRIVATE* if the acquirer privatized the target post-acquisition, *COLLAR* for transactions with a collar structure, *TERM* for deals with termination fees, *FRIENDLY* if the deal attitude is considered to be friendly, and *US* if the bidder is a U.S.-based company. *PREM1D* refers to the premium of the offer price over the target's closing stock price one day prior to the original announcement date, expressed as a percentage. *PRICE* denotes the price per common share paid by the acquirer, *TRUNUP* the target's pre-announcement cumulative abnormal stock return, *TANNRET* the target's announcement-day abnormal return, *TTPRET1* the target's post-announcement cumulative abnormal return. *ARUNUP* is the acquirer's pre-announcement abnormal stock return, and *MKTVOL* defines the market volume on the day before the announcement. *ABNORMVOLC* is the total abnormal call volume for the target over the 30 days preceding the announcement. All specifications have year fixed effects. We report the number of observations (*Observations*) and the pseudo *R*-squared (ps.R²). ***, ** and * denote statistical significance at the 1%, 5% and 10% levels, based on Firth's method for bias reduction in logistic regressions.

VARIABLES	(1) Logit (Odds Ratio)	(2) Logit (Odds Ratio)	(3) Logit (Odds Ratio)	(4) Logit (Odds Ratio)
SIZE	0.63*** (1.87)	0.44* (1.55)	0.79*** (2.21)	0.79*** (2.20)
CASH	0.15 (1.17)	0.10 (1.10)	0.02 (1.02)	0.01 (1.01)
CHALLENGE	-0.64 (0.53)	-0.76 (0.47)	-0.57 (0.56)	-0.59 (0.55)
COMPLETE	1.05* (2.87)	1.06* (2.88)	1.07* (2.92)	1.07* (2.92)
TOE	-0.76 (0.47)	-0.74 (0.48)	-0.73 (0.48)	-0.73 (0.48)
PRIVATE	0.20 (1.22)	0.27 (1.31)	0.30 (1.35)	0.31 (1.36)
COLLAR	0.43 (1.53)	0.37 (1.45)	0.30 (1.35)	0.29 (1.33)
TERM	0.67 (1.95)	0.60 (1.83)	0.63 (1.88)	0.64 (1.89)
FRIENDLY	-0.36 (0.70)	-0.36 (0.70)	-0.34 (0.71)	-0.35 (0.70)
US	-0.55** (0.58)	-0.59** (0.55)	-0.57** (0.57)	-0.56** (0.57)
PREM1D		0.01*** (1.01)		
PRICE		0.01*** (1.01)		
TRUNUP			-0.65 (0.52)	-0.68 (0.51)
TANNRET			-0.69 (0.50)	-0.67 (0.51)
TTPRET1			2.10*** (8.18)	2.12*** (8.31)
ARUNUP			0.12 (1.12)	0.07 (1.07)
MKTVOL				0.00 (1.00)
ABNORMVOLC				0.00 (1.00)
Constant	-3.56*** (0.03)	-3.85*** (0.02)	-3.78*** (0.02)	-3.86*** (0.02)
Observations	1,859	1,807	1,859	1,859
ps.R ²	0.11	0.12	0.13	0.16

Figure 1: Abnormal Trading Volumes before Announcement Dates

In Figure (1a), we depict the average abnormal trading volume for all equity options (solid line), call options (dashed line) and put options (dotted line), respectively, for the target companies, over the 30-day pre-announcement period. Volume is defined as the number of option contracts. In Figure (1b), we depict the average cumulative abnormal trading volume for all options (solid line), call options (dashed line) and put options (dotted line) over the same event period. Statistics are computed for a sample of 1,859 target companies over the time period January 1996 through December 31, 2012.

(a)



(b)

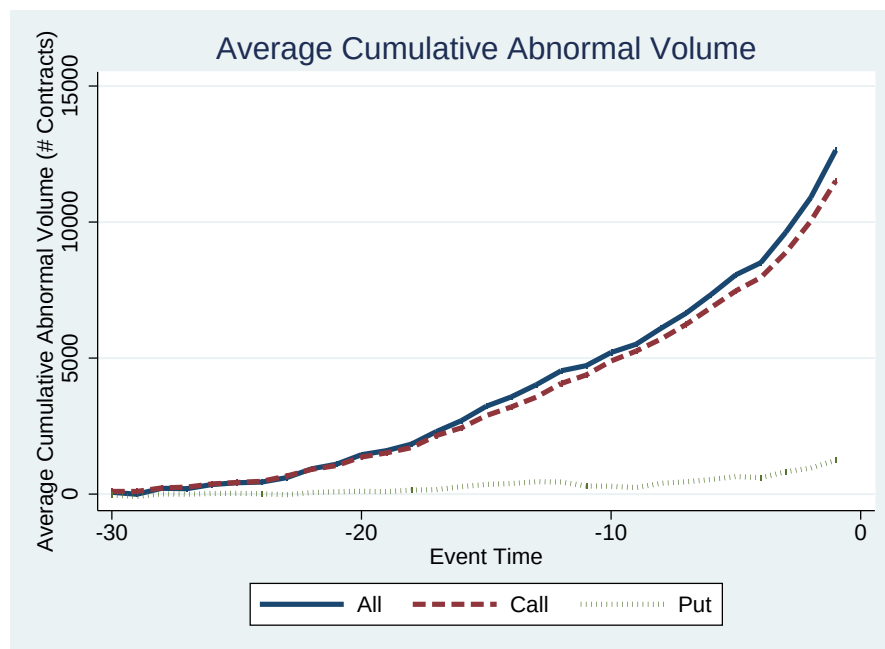


Figure 2: Abnormal Trading Volumes - Treatment and Matched Control Groups, News and Rumors

In Figures (2a) and (2b), we plot the average and average cumulative abnormal trading volume, respectively, for aggregate options volume in the treatment group (Main - dashed line) and the propensity-matched control group using the best match (PS1 - solid line), over the 30-day pre-announcement period. Volume is defined as the number of option contracts, and abnormal volume is based on the market model for volume (MMV). Firms are matched based on 3-month moving average values in the month preceding the announcement for the target's stock return, its percentage bid-ask spread, the stock trading volume, and stock return volatility using an EWMA model with an autoregressive coefficient of 0.94. In Figures (2c) and (2d), we plot, respectively, the average and average cumulative abnormal trading volume for aggregate options volume in the sample with (No News – solid line) and without (News (30d) – dashed line) news or rumors about takeovers in the 30 pre-announcement days. These results are based on a log-transformation of volume, defined as $\log Volume = \ln(1 + Volume)$. All statistics are computed for a sample of 1,859 target companies over the time period January 1996 through December 31, 2012.

